

REBRIKHA, Russian Federation

WMO: 299230

Lat: 53.082N

Lon: 82.338E

Elev: 719

StdP: 14.32

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.3	(c) -28.6	(d) -39.4	(e) 0.6	(f) -33.9	(g) -33.8	(h) 0.8	(i) -28.4	(j) 31.3	(k) 17.8	(l) 27.3	(m) 19.0	(n) 1.4	(o) 250	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	87.0	65.0	83.6	64.1	80.7	63.2	68.9	79.9	67.3	78.2	65.7	76.5	8.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	96.5	73.5	63.6	90.3	71.7	61.7	84.4	70.3	33.6	79.8	32.2	78.2	31.0	76.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	22.2	19.0	-39.9	92.8	6.9	3.7	-44.9	95.4	-49.0	97.6	-52.8	99.7	-57.9	102.3
			-39.8	71.4	6.3	2.3	-44.3	73.1	-48.0	74.5	-51.5	75.8	-56.1	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	0.3	5.2	19.8	39.4	53.6	64.1	66.8	62.6	51.1	38.5	20.8	7.2
	DBStd	25.92	15.84	14.24	12.29	10.36	9.43	7.12	5.91	6.61	8.03	9.33	14.70	14.84
	HDD50	6809	1542	1255	937	341	74	3	0	2	82	369	876	1328
	HDD65	10854	2007	1675	1402	768	366	99	50	125	422	821	1326	1793
	CDD50	1678	0	0	0	24	186	427	522	391	115	13	0	0
	CDD65	247	0	0	0	0	14	73	107	49	4	0	0	0
	CDH74	2920	0	0	0	14	294	833	1006	655	116	2	0	0
	CDH80	789	0	0	0	1	74	221	284	181	28	0	0	0
(14) Wind	WSAvg	7.2	7.4	7.5	7.9	7.9	7.6	6.1	5.0	5.2	5.9	7.7	8.8	9.0
(15) Precipitation	PrecAvg	16.0	0.7	0.6	0.8	0.9	1.5	2.0	2.8	1.9	1.3	1.3	1.4	1.0
	PrecMax	22.8	1.5	1.7	1.9	3.6	4.4	4.9	6.0	5.5	3.0	2.7	2.7	2.2
	PrecMin	9.2	0.0	0.0	0.2	0.0	0.3	0.6	0.6	0.4	0.2	0.5	0.5	0.4
	PrecStd	3.7	0.4	0.4	0.5	0.7	0.9	1.1	1.4	1.1	0.7	0.6	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	35.5	48.9	75.8	87.3	89.4	90.8	90.9	83.8	70.3	52.5	35.6
		MCWB	31.9	32.1	40.1	54.1	60.9	64.0	67.9	66.6	59.2	52.2	43.7	32.6
	2%	DB	30.3	31.9	42.2	68.3	81.3	85.5	86.6	84.9	76.9	63.8	47.1	32.6
		MCWB	28.3	29.5	36.5	52.0	58.5	63.7	67.0	64.8	57.6	49.6	40.7	30.5
	5%	DB	26.3	28.7	38.9	62.8	77.0	82.2	83.0	81.0	71.4	58.4	41.8	29.4
		MCWB	24.2	26.2	34.6	48.8	56.6	63.3	66.0	63.5	55.8	46.9	37.1	27.4
	10%	DB	22.3	25.1	35.6	56.7	72.1	78.9	79.8	77.4	66.9	53.0	37.5	26.5
		MCWB	20.7	23.0	32.3	45.8	55.0	62.3	65.3	62.4	53.9	44.5	34.5	24.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.1	40.7	56.6	63.5	69.0	72.0	70.2	62.8	54.0	45.6	33.5
		MCDB	34.3	34.7	45.9	71.9	80.8	81.1	83.3	81.2	76.1	65.9	49.6	34.6
	2%	WB	28.5	29.7	37.2	52.9	60.6	67.0	69.8	67.4	59.4	50.6	40.5	30.8
		MCDB	30.3	31.8	40.9	65.1	75.6	79.5	80.6	79.9	72.1	61.3	46.2	32.6
	5%	WB	24.4	26.4	34.8	49.7	58.4	65.5	68.2	65.5	57.2	47.9	37.3	27.6
		MCDB	26.2	28.5	38.2	61.1	72.9	77.5	78.3	77.8	69.5	57.0	41.0	29.1
	10%	WB	20.7	23.1	32.7	46.8	56.3	63.8	66.7	63.6	54.9	45.4	34.3	24.8
		MCDB	22.3	25.1	35.5	56.5	69.6	75.5	76.3	74.9	65.4	52.2	37.4	26.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	19.0	18.4	19.7	23.0	20.8	19.5	22.0	22.1	17.3	13.0	15.2
		MCDBR	14.8	15.8	18.0	30.4	31.3	27.4	24.9	28.9	31.8	29.2	15.4	14.8
	5% WB	MCWBR	13.9	14.2	13.5	16.1	13.2	10.9	9.2	11.9	15.0	16.4	10.9	13.9
		MCWBR	15.3	15.6	15.7	27.3	25.5	21.7	20.2	24.4	27.4	26.3	13.9	14.8
(40) Clear-Sky Solar Irradiance	taub		0.296	0.332	0.384	0.410	0.411	0.408	0.431	0.409	0.371	0.355	0.338	0.303
		taud	2.022	1.958	1.905	2.117	2.192	2.284	2.215	2.254	2.305	2.232	2.050	1.967
	Ebn at Noon		201	231	244	259	266	268	258	257	253	226	181	167
		Edh at Noon	25	36	48	44	43	40	42	38	32	28	24	22
(44) All-Sky Solar Radiation	RadAvg		358	697	1145	1480	1802	1991	1798	1550	1106	581	296	245
	RadStd		29	47	108	92	163	139	109	103	104	78	43	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page